Supplementary Material

Physics-Guided Probabilistic Glucose Forecasting in Type 1 Diabetes: An Audited Multi-Horizon Evaluation with Validated Patient-Specific Insulin Sensitivity

Notation, Variables, and Terminology

Rate constants are expressed per minute unless stated otherwise. A subscript b denotes a basal or fasting quantity, a hat denotes a prediction, bold lowercase symbols denote vectors, and bold uppercase symbols denote matrices. Bracketed reference numbers refer to the reference list of the main text. The patient-parameter vector θ_p is listed in Table XIV of the main text.

Table S1: Physiological state variables and external inputs

Symbol	Unit	Name	Meaning in this study
$G(t)$	mg/dL	Plasma glucose	Glucose concentration forecast by the learned spline and constrained by the mechanistic equation.
G_b	mg/dL	Basal glucose	Subject-specific fasting anchor used by the Bergman equilibrium.
G_0	mg/dL	Anchor glucose	Last real CGM observation at forecast time $t = 0$.
$X(t)$	min^{-1}	Remote insulin action	Interstitial insulin effect entering glucose disposal additively with p_1.
$I(t)$	μU/mL	Plasma insulin	Circulating insulin concentration generated by subcutaneous absorption and clearance.
I_b	μU/mL	Basal plasma insulin	Concentration implied by the subject's basal delivery; subtracting it preserves basal equilibrium.
S_1, S_2	U	SC insulin states	First and second subcutaneous absorption compartments.
Q_{sto}	g	Stomach carbohydrate	Ingested carbohydrate awaiting gastric emptying.
Q_{gut}	g	Gut carbohydrate	Carbohydrate available for intestinal absorption.
$R_a(t)$	mg/min	Glucose appearance rate	Meal-derived glucose entering the accessible glucose compartment.
u_{ins}	U/min	Insulin input	Basal plus bolus insulin delivery on the five-minute grid.
u_{carb}	g/min	Meal input	Recorded carbohydrate ingestion on the grid.
$\mathbf{z}$	–	Compartment state	$[S_1, S_2, I, Q_{\text{sto}}, Q_{\text{gut}}, X]^\mathsf{T}$.
$\mathbf{u}$	–	Input vector	$[u_{\text{ins}}, u_{\text{carb}}, I_b]^\mathsf{T}$.

Table S2: Architecture, discretization, and uncertainty notation

Symbol	Value or dimension	Definition
t	0–120 min	Time measured from the forecast anchor.
h	30, 60, 90, 120 min	Forecast horizon.
Δ	5 min	Common temporal grid step.
B	batch dependent	Number of windows processed together.
H	4	Number of reported forecast horizons.
K	12	Number of cubic B-spline basis functions.
$B_k(t)$	spline basis	The kth clamped, uniformly knotted B-spline basis function.
c_k	learned scalar	Spline coefficient emitted by the trajectory head.
$G_\theta(t)$	mg/dL	Anchored learned trajectory $G_0 + \sum_k c_k[B_k(t) - B_k(0)]$.
A, B	6×6, 6×3	Continuous compartment-state and input matrices.
A_d, B_d	discrete matrices	Exact zero-order-hold matrices obtained from one augmented matrix exponential.
θ_p	10 elements	Patient-specific physiological parameter vector in Table XIV of the main text.
$g = \sigma(\gamma)$	initialized near 0.10	Learned trust gate controlling the mechanistic prior.
$\hat{G}_{0.5}$	mg/dL	Median forecast and reported point prediction.
$\hat{G}_q$	mg/dL	Forecast at quantile level $q \in \{0.1, 0.5, 0.9\}$.
$\delta_q(h)$	mg/dL	Positive softplus-derived distance from the median, preventing quantile crossing.
$r(t)$	dimensionless	Scale-normalized physics residual evaluated at 121 collocation points.
n_{eff}	$\approx N/48$	Approximate independent sample count after accounting for window overlap.

Table S3: Training-loss terminology

Term	Meaning
$\mathcal{L}_{\text{Huber}}$	Horizon-weighted robust point loss for the median forecast; quadratic near zero error and linear for large errors.
$\mathcal{L}_{\text{pinball}}$	Proper quantile loss $\max\{q(y-\hat{y}_q),(q-1)(y-\hat{y}_q)\}$.
$\mathcal{L}_{\text{phys}}$	Mean squared, nondimensionalized Bergman residual over the collocation grid.
$\mathcal{L}_{\text{prior}}$	Weak population-prior penalty that discourages physiologically implausible parameter drift.
$\mathcal{L}_{\text{temporal}}$	Within-subject penalty on disagreement between parameter estimates from adjacent windows.
λ_{phys}	Physics weight, adaptive in the relevant ablation arms and optionally multiplied by a curriculum ramp.
$s_i = \log \sigma_i^2$	Stable log-variance parameterization used for learned multi-objective weighting [51].

Table S4: Evaluation metrics and clinical terms

Term	Unit	Definition and interpretation
MAE	mg/dL	Mean absolute error, $n^{-1}\sum_i \lvert\hat{y}_i - y_i\rvert$.
RMSE	mg/dL	Root-mean-square error, emphasizing larger deviations.
R^2	–	$1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$.
MARD	%	Mean absolute relative difference, with reference glucose in the denominator.
Skill	%	$1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{persistence}}$. Positive values beat no-change prediction.
TIR/TAR/TBR	%	Time in, above, or below the consensus glucose range. The main in-range interval is 70–180 mg/dL [55].
CV ratio	–	$\text{CV}_{\text{pred}}/\text{CV}_{\text{actual}}$; values below one quantify excursion compression.
LBGI/HBGI	–	Low/high blood-glucose risk indices after the Kovatchev symmetric risk transform.
Coverage	–	Fraction of observations below a stated predictive quantile; calibration requires observed and nominal coverage to agree.
Hypo sensitivity	–	Fraction of actual $G < 70$ mg/dL events detected by the forecast or alarm.
Precision	–	Fraction of hypoglycemia alarms corresponding to actual events.
ICC(1)	–	One-way, single-measure intraclass correlation used for S_I test–retest stability [60].
ρ	–	Spearman rank correlation used for monotonic association with insulin dose [61].
IG_i	mg/dL	Integrated-gradient attribution of feature i relative to the standardized-mean baseline.
Clarke/Parkes EGA	zone	Point-wise clinical error grids originally designed for glucose-measurement error.
PRED-EGA	category	Prediction-specific error grid combining level and rate information [69]; reported in the Results of the main text.

Table S5: Abbreviations and protocol terminology

Short form	Expansion	Study-specific meaning
T1D/IDDM	Type 1 diabetes / insulin-dependent diabetes mellitus	IDDM is retained only when describing older physiological literature.
CGM	Continuous glucose monitoring	Nominal five-minute glucose sampling in OhioT1DM.
SC	Subcutaneous	Route represented by insulin compartments S_1 and S_2.
IOB/COB	Insulin/carbohydrate on board	Derived as $S_1 + S_2$ and $Q_{\text{sto}} + Q_{\text{gut}}$.
TDD/ICR	Total daily insulin dose / insulin-to-carbohydrate ratio	External consistency variables for the physiology analysis; ICR is available only for six subjects.
PINN/ODE	Physics-informed neural network / ordinary differential equation	Neural training constrained by the physiological differential equations.
LOSO	Leave-one-subject-out	Twelve subject-disjoint folds, each withholding all data from one test subject.
Official protocol	OhioT1DM temporal holdout	Later data from subjects whose earlier data are available for training; therefore personalized.
BGLP	Blood Glucose Level Prediction challenge	OhioT1DM challenge setting; 2018 and 2020 have different test warmup rules.
EGA	Error-grid analysis	Clarke and Parkes clinical zone classification.
IG/SHAP	Integrated gradients / SHapley Additive exPlanations	IG is used; the legacy single-timestep SHAP procedure was rejected.
SD/CI/ICC	Standard deviation / confidence interval / intraclass correlation	Uncertainty and reliability summaries computed at the subject level.
A0–A7	Ablation arms	Controlled variants of physics form, weighting, parameter personalization, and curriculum.
ZOH	Zero-order hold	Exact interval propagation under piecewise-constant inputs; also used in the literature for persistence.